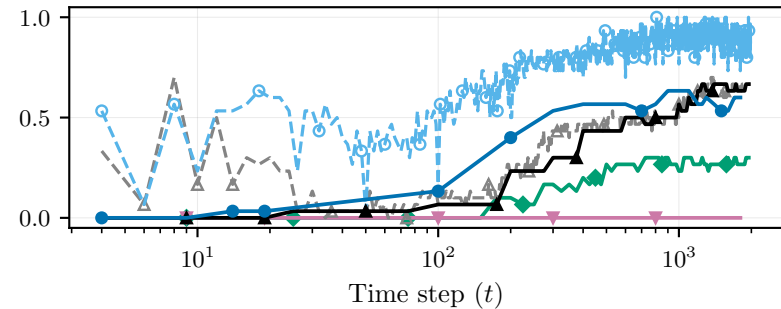


Confidence $1 - \hat{\beta}_t$ 

—●— online clustering

—▲— DRO

—◆— SAA

... upper bound

-○- online clustering subgrad

-▲- DRO subgrad

—▼— cluster SAA

—●— empirical regret

Online clustering dynamic regret

